Physics 210 Tutorial 5 Electric Potential

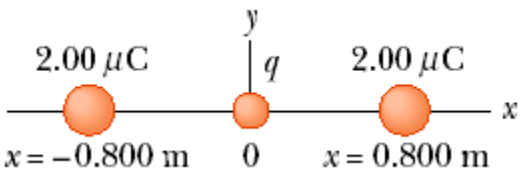
These problems are from an old text. The numbers match the numbers in the solutions.

3. (a) Calculate the speed of a proton that is accelerated from rest through a potential difference of 120 V. (b) Calculate the speed of an electron that is accelerated through the same potential difference.

37. The potential in a region between $x = 0$ and $x = 6.00$ m is $V = a + bx$, where $a = 10.0$ V and $b = -7.00$ V/m. Determine (a) the potential at $x = 0$, 3.00 m, and 6.00 m, and (b) the magnitude and direction of the electric field at $x = 0$, 3.00 m, and 6.00 m.

Note: Unless stated otherwise, assume the reference level of potential is $V = 0$ at $r = \infty$.

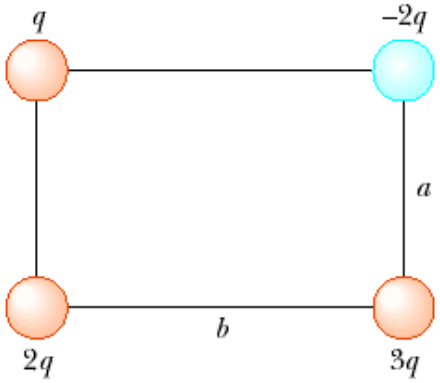
16. Given two 2.00-μC charges, as shown in the figure below, and a positive test charge $q = 1.28 \times 10^{-18}$ C at the origin, (a) what is the net force exerted by the two 2.00-μC charges on the test charge q? (b) What is the electric field at the origin due to the two 2.00-μC charges? (c) What is the electrical potential at the origin due to the two 2.00-μC charges?



17. At a certain distance from a point charge, the magnitude of the electric field is 500 V/m and the electric potential is –3.00 kV. (a) What is the distance to the charge? (b) What is the magnitude of the charge?

18. A charge $+q$ is at the origin. A charge $-2q$ is at $x = 2.00$ m on the x axis. For what finite value(s) of x is (a) the electric field zero? (b) the electric potential zero?

34. Calculate the energy required to assemble the array of charges shown in the figure below, where $a = 0.200$ m, $b = 0.400$ m, and $q = 6.00\ \mu$C.



39. Over a certain region of space, the electric potential is $V = 5x - 3x^2y + 2yz^2$. Find the expressions for the x, y, and z components of the electric field over this region. What is the magnitude of the field at the point P that has coordinates (1, 0, –2) m?

42. Consider a ring of radius R with the total charge Q spread uniformly over its perimeter. What is the potential difference between the point at the center of the ring and a point on its axis a distance $2R$ from the center?

63. From Gauss's law, the electric field set up by a uniform line of charge is

$$E = \lambda/2\pi\varepsilon_0 r$$

where λ is the linear charge density along the line. Derive an expression for the potential difference between $r = r_1$ and $r = r_2$.